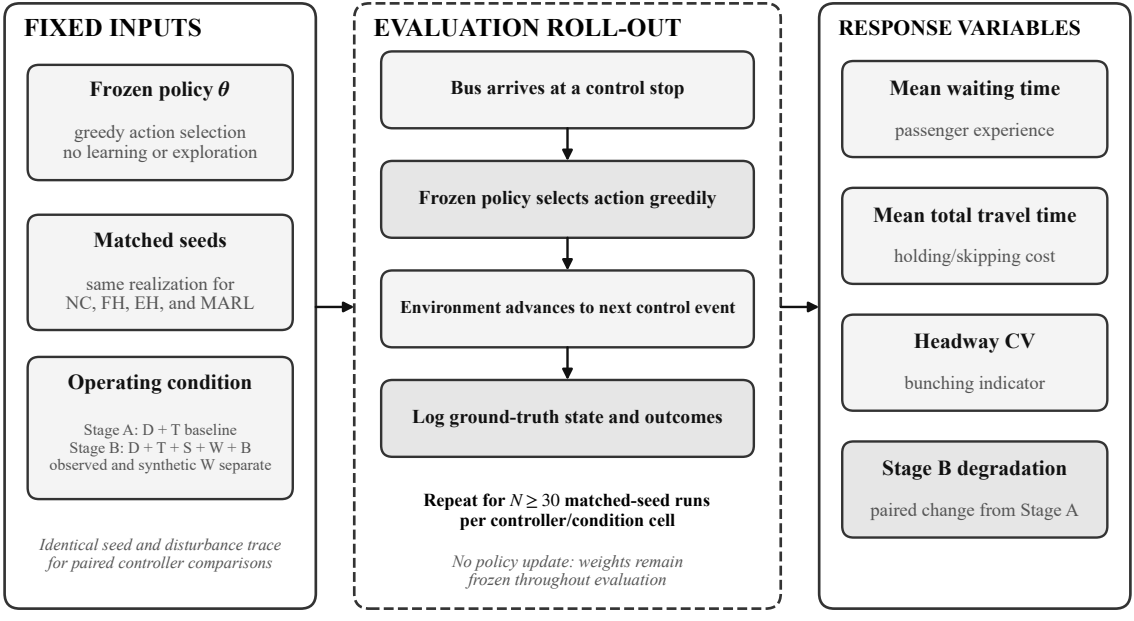

FIXED INPUTS
Frozen policy θ
greedy action selection
no learning or exploration
Matched seeds
same realization for
NC, FH, EH, and MARL
Operating condition
Stage A: D + T baseline
Stage B: D + T + S + W + B
observed and synthetic W separate
Identical seed and disturbance trace
for paired controller comparisons
EVALUATION ROLL-OUT
Bus arrives at a control stop
Frozen policy selects action greedily
Environment advances to next control event
Log ground-truth state and outcomes
Repeat for $N \geq 30$ matched-seed runs
per controller/condition cell
No policy update: weights remain
frozen throughout evaluation
RESPONSE VARIABLES
Mean waiting time
passenger experience
Mean total travel time
holding/skipping cost
Headway CV
bunching indicator
Stage B degradation
paired change from Stage A